```
# importing required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
from sklearn.cluster import KMeans


# reading the data and looking at the first five rows of the data
data=pd.read_csv("Wholesale customers data.csv")
data.head()
```

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen
0	2	3	12669	9656	7561	214	2674	1338
1	2	3	7057	9810	9568	1762	3293	1776
2	2	3	6353	8808	7684	2405	3516	7844
3	1	3	13265	1196	4221	6404	507	1788
4	2	3	22615	5410	7198	3915	1777	5185

```
# statistics of the data
data.describe()
```

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen
count	440.000000	440.000000	440.000000	440.000000	440.000000	440.000000	440.000000	440.000000
mean	1.322727	2.543182	12000.297727	5796.265909	7951.277273	3071.931818	2881.493182	1524.870455
std	0.468052	0.774272	12647.328865	7380.377175	9503.162829	4854.673333	4767.854448	2820.105937
min	1.000000	1.000000	3.000000	55.000000	3.000000	25.000000	3.000000	3.000000
25%	1.000000	2.000000	3127.750000	1533.000000	2153.000000	742.250000	256.750000	408.250000
50%	1.000000	3.000000	8504.000000	3627.000000	4755.500000	1526.000000	816.500000	965.500000
75%	2.000000	3.000000	16933.750000	7190.250000	10655.750000	3554.250000	3922.000000	1820.250000
max	2.000000	3.000000	112151.000000	73498.000000	92780.000000	60869.000000	40827.000000	47943.000000

There is a lot of variation in the magnitude of the data. Variables like Channel and Region have low magnitude whereas variables like Fresh, Milk, Grocery, etc. have a higher magnitude.

Since K-Means is a distance-based algorithm, this difference of magnitude can create a problem. So let’s first bring all the variables to the same magnitude:

```
# standardizing the data
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
data_scaled = scaler.fit_transform(data)

# statistics of scaled data
pd.DataFrame(data_scaled).describe()
```

	0	1	2	3	4	5	6	7
count	4.400000e+02	4.400000e+02	4.400000e+02	4.400000e+02	4.400000e+02	4.400000e+02	4.400000e+02	4.400000e+02
mean	-2.452584e-16	-5.737834e-16	-2.422305e-17	-1.589638e-17	-6.030530e-17	1.135455e-17	-1.917658e-17	-8.276208e-17
std	1.001138e+00	1.001138e+00	1.001138e+00	1.001138e+00	1.001138e+00	1.001138e+00	1.001138e+00	1.001138e+00
min	-6.902971e-01	-1.995342e+00	-9.496831e-01	-7.787951e-01	-8.373344e-01	-6.283430e-01	-6.044165e-01	-5.402644e-01
25%	-6.902971e-01	-7.023369e-01	-7.023339e-01	-5.783063e-01	-6.108364e-01	-4.804306e-01	-5.511349e-01	-3.964005e-01
50%	-6.902971e-01	5.906683e-01	-2.767602e-01	-2.942580e-01	-3.366684e-01	-3.188045e-01	-4.336004e-01	-1.985766e-01
75%	1.448652e+00	5.906683e-01	3.905226e-01	1.890921e-01	2.849105e-01	9.946441e-02	2.184822e-01	1.048598e-01
max	1.448652e+00	5.906683e-01	7.927738e+00	9.183650e+00	8.936528e+00	1.191900e+01	7.967672e+00	1.647845e+01

```
# defining the kmeans function with initialization as k-means++
kmeans = KMeans(n_clusters=2, init='k-means++')

# fitting the k means algorithm on scaled data
kmeans.fit(data_scaled)
```

```
KMeans(n_clusters=2)
```

```
# inertia on the fitted data
kmeans.inertia_
```

```
2599.3855593561393
```

```
# fitting multiple k-means algorithms and storing the values in an empty list
SSE = []
for cluster in range(1,20):
    kmeans = KMeans(n_jobs = -1, n_clusters = cluster, init='k-means++')
    kmeans.fit(data_scaled)
    SSE.append(kmeans.inertia_)

# converting the results into a dataframe and plotting them
frame = pd.DataFrame({'Cluster':range(1,20), 'SSE':SSE})
plt.figure(figsize=(12,6))
plt.plot(frame['Cluster'], frame['SSE'], marker='o')
plt.xlabel('Number of clusters')
plt.ylabel('Inertia')
```

```
C:\Users\Nouman\anaconda3\lib\site-packages\sklearn\cluster\_kmeans.py:938: FutureWarning: 'n_jobs' was deprecated in version 0.23 and
  warnings.warn("'n_jobs' was deprecated in version 0.23 and will be"
C:\Users\Nouman\anaconda3\lib\site-packages\sklearn\cluster\_kmeans.py:938: FutureWarning: 'n_jobs' was deprecated in version 0.23 and
  warnings.warn("'n_jobs' was deprecated in version 0.23 and will be"
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  warnings.warn("'n_jobs' was deprecated in version 0.23 and will be"
Text(0, 0.5, 'Inertia')
```
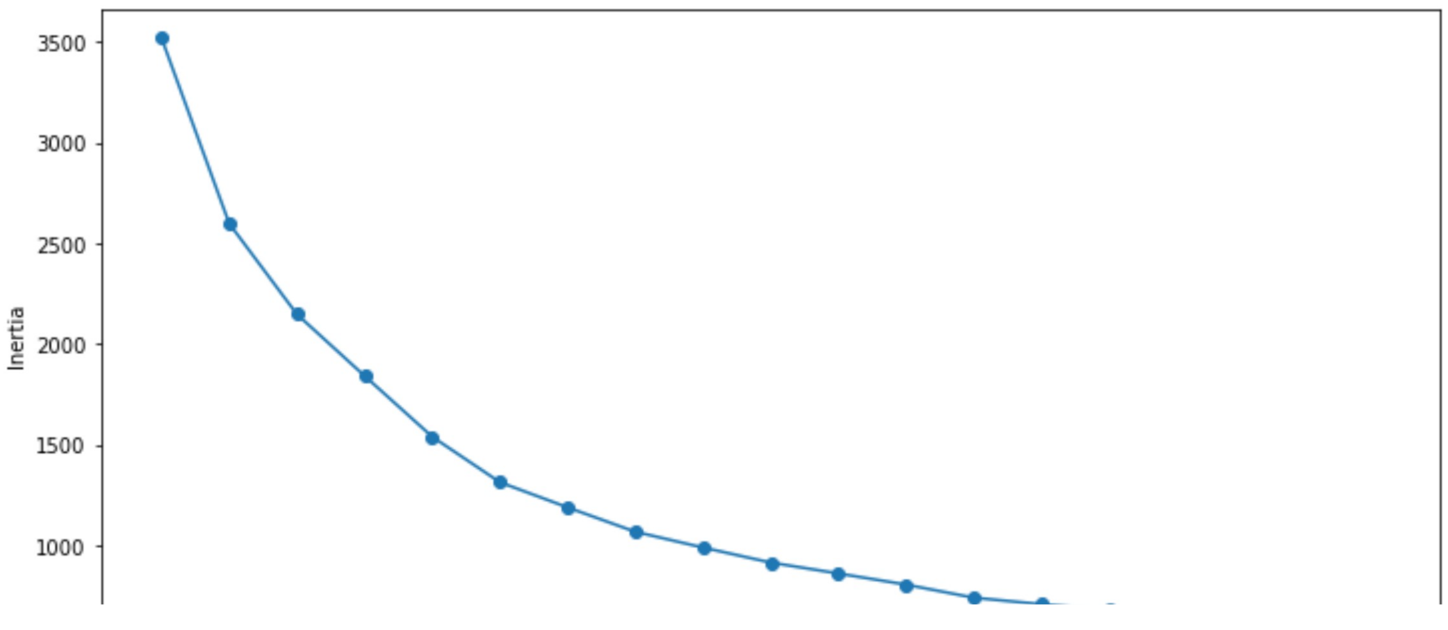
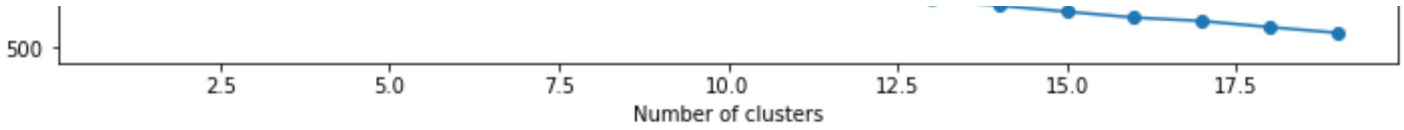






```
# k means using 5 clusters and k-means++ initialization
kmeans = KMeans(n_clusters = 5, init='k-means++')
kmeans.fit(data_scaled)
pred = kmeans.predict(data_scaled)
pred
```

```
array([4, 4, 4, 1, 4, 4, 4, 4, 1, 4, 4, 4, 4, 4, 4, 1, 4, 1, 4, 1, 4, 1,
       1, 2, 4, 4, 1, 1, 4, 1, 1, 1, 1, 1, 1, 4, 1, 4, 4, 1, 1, 1, 4, 4,
       4, 4, 4, 2, 4, 4, 1, 1, 4, 4, 1, 1, 2, 4, 1, 1, 4, 2, 4, 4, 1, 2,
       1, 4, 1, 1, 1, 1, 1, 4, 4, 1, 1, 4, 1, 1, 1, 4, 4, 1, 4, 2, 2, 1,
       1, 1, 1, 1, 2, 1, 4, 1, 4, 1, 1, 1, 4, 4, 4, 1, 1, 1, 4, 4, 4, 4,
       1, 4, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 4, 1, 1, 1, 4, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 4, 1, 1, 1, 1, 1, 1, 1, 1,
       1, 4, 4, 1, 4, 4, 4, 1, 1, 4, 4, 4, 4, 1, 1, 1, 4, 4, 1, 4, 1, 4,
       1, 1, 1, 1, 1, 2, 1, 3, 1, 1, 1, 1, 4, 4, 1, 1, 1, 4, 1, 1, 0, 4,
       0, 0, 4, 4, 0, 0, 0, 4, 0, 0, 0, 4, 0, 2, 0, 0, 4, 0, 4, 0, 4, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 4, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       4, 0, 4, 0, 4, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 4, 1, 4, 1, 1, 1, 1,
       1, 1, 1, 1, 1, 1, 1, 4, 0, 4, 0, 4, 4, 0, 4, 4, 4, 4, 4, 4, 4, 0,
       0, 4, 0, 0, 4, 0, 0, 4, 0, 0, 0, 4, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0,
       0, 4, 0, 2, 0, 4, 0, 0, 0, 0, 4, 4, 1, 4, 1, 1, 4, 4, 1, 4, 1, 4,
       1, 4, 1, 1, 1, 4, 1, 1, 1, 1, 1, 1, 1, 4, 1, 1, 1, 1, 4, 1, 1, 4,
       1, 1, 4, 1, 1, 4, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
       4, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 4, 4, 1, 1, 1, 1, 1, 1, 4, 4, 1,
       4, 1, 1, 4, 1, 4, 4, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 4, 1, 1])
```

```
# k means using 5 clusters and k-means++ initialization
kmeans = KMeans(n_clusters = 5, init='k-means++')
kmeans.fit(data_scaled)
pred = kmeans.predict(data_scaled)
```

```
import matplotlib.pyplot as plt
#from kneed import KneeLocator
from sklearn.datasets import make_blobs
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.preprocessing import StandardScaler
```

We can generate the data from the above GIF using make_blobs(), a convenience function in scikit-learn used to generate synthetic clusters. make_blobs() uses these parameters:

- n_samples is the total number of samples to generate.
- centers is the number of centers to generate.
- cluster_std is the standard deviation.

make_blobs() returns a tuple of two values:

- A two-dimensional NumPy array with the x- and y-values for each of the samples
- A one-dimensional NumPy array containing the cluster labels for each sample

```
features, true_labels = make_blobs(
    n_samples=200,
    centers=3,
    cluster_std=2.75,
    random_state=42
)
```

```
features[:5]
```

```
array([[  9.77075874,   3.27621022],
       [ -9.71349666,  11.27451802],
       [ -6.91330582,  -9.34755911],
       [-10.86185913, -10.75063497],
       [ -8.50038027,  -4.54370383]])
```

```
true_labels[:5]
```

```
     array([1, 0, 2, 2, 2])
```

```
scaler = StandardScaler()
scaled_features = scaler.fit_transform(features)
```

```
scaled_features[:5]
```

```
     array([[ 2.13082109,  0.25604351],
            [-1.52698523,  1.41036744],
            [-1.00130152, -1.56583175],
            [-1.74256891, -1.76832509],
            [-1.29924521, -0.87253446]])
```

```
kmeans = KMeans(
    init="random",
    n_clusters=3,
    n_init=10,
    max_iter=300,
    random_state=42
)
```

```
kmeans.fit(scaled_features)
```

```
     KMeans(init='random', n_clusters=3, random_state=42)
```

```
# The lowest SSE value
kmeans.inertia_
```

```
     74.57960106819854
```

```
# Final locations of the centroid
kmeans.cluster_centers_
```

```
     array([[ 1.19539276,  0.13158148],
            [-0.25813925,  1.05589975],
            [-0.91941183, -1.18551732]])
```

```
# The number of iterations required to converge
kmeans.n_iter_
```

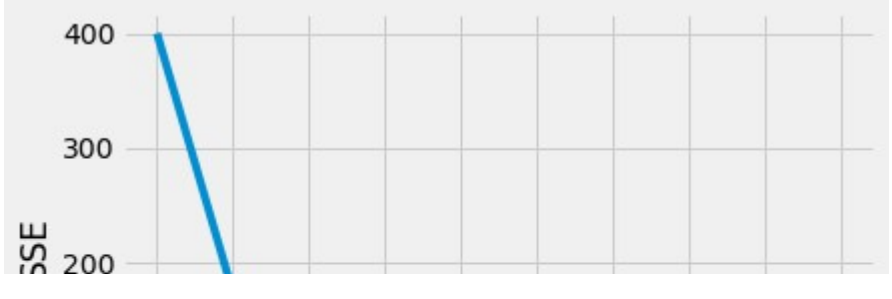
```
     6
```

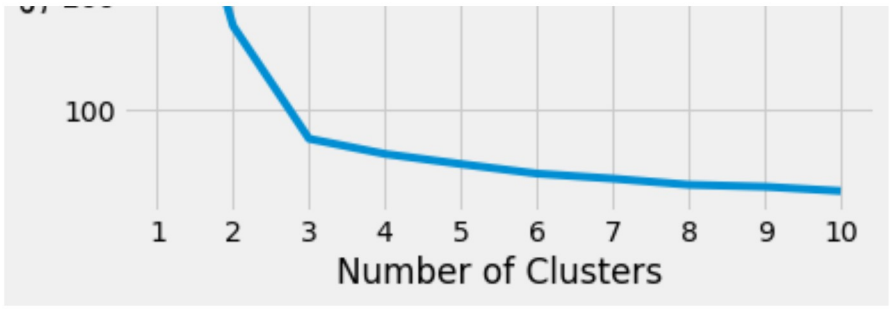
```
kmeans.labels_[:5]
```

```
     array([0, 1, 2, 2, 2])
```

```
kmeans_kwargs = {
    "init": "random",
    "n_init": 10,
    "max_iter": 300,
    "random_state": 42,
}

# A list holds the SSE values for each k
sse = []
for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, **kmeans_kwargs)
    kmeans.fit(scaled_features)
    sse.append(kmeans.inertia_)
```

```
 plt.style.use("fivethirtyeight")
 plt.plot(range(1, 11), sse)
 plt.xticks(range(1, 11))
 plt.xlabel("Number of Clusters")
 plt.ylabel("SSE")
 plt.show()
```





Start coding or generate with AI.